

“Feariosity”-Guided Reinforcement Learning for Safe and Efficient Autonomous End-to-End Navigation

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Abstract—End-to-end navigation strategies using reinforcement learning (RL) can improve the adaptability and autonomy of Autonomous ground vehicles (AGVs) in complex environments. However, RL still faces challenges in data efficiency and safety. Neuroscientific and psychological research shows that during exploration, the brain balances between fear and curiosity, a critical process for survival and adaptation in dangerous environments. Inspired by this scientific insight, we propose the “Feariosity” model, which integrates fear and curiosity model to simulate the complex psychological dynamics organisms experience during exploration. Based on this model, we developed an innovative policy constraint method that evaluates potential hazards and applies necessary safety constraints while encouraging exploration of unknown areas. Additionally, we designed a new experience replay mechanism that quantifies the threat and unknown level of data, optimizing their usage probability. Extensive experiments in both simulation and real-world scenarios demonstrate that the proposed method significantly improves data efficiency, asymptotic performance during training. Furthermore, it achieves higher success rates, driving efficiency, and robustness in deployment. This also highlights the key role of mimicking biological neural and psychological mechanisms in improving the safety and efficiency through RL.

Index Terms—Autonomous navigation, fear and curiosity, safety constraints, experience replay optimization.

I. INTRODUCTION

AUTONOMOUS ground vehicles (AGVs) are transforming robotics and intelligent transportation by reducing human workload and enabling autonomous mobility [1]. While end-to-end navigation approaches based on neural networks (NNs) have shown promise for complex driving scenarios [2], imitation learning (IL) often struggles to generalize beyond the

training data [3]. Consequently, reinforcement learning (RL) is increasingly adopted, as it allows agents to learn effective navigation policies through trial-and-error in interactive environments [4]. RL has achieved notable progress in robotics and transportation [5], [6], yet two persistent challenges remain: limited data efficiency during training and insufficient safety during deployment [7].

RL is closely related to the nervous system, as both rely on learning through reward and punishment mechanisms [8]. The RL update process simulates the dopamine and striatum learning mechanism, while NNs mimic the brain’s information processing [9]. Recent research highlights the amygdala’s role in anticipating danger, crucial for survival and adaptation in hazardous environments [10]. Studies in neuroscience and psychology suggest that fear helps avoid threats but can cause hesitation, reducing exploration, while curiosity drives exploration and enhances understanding of unfamiliar environments [11]. However, most RL-based navigation frameworks treat these aspects separately or fail to balance them, leading to overly conservative or risk-prone policies.

Motivated by these insights, we propose a novel RL framework for AGV navigation that explicitly models and balances fear and curiosity. Our approach constructs a unified “Feariosity” model, leveraging a feature extractor and a learned world model to evaluate both the risk (fear) and novelty (curiosity) of state-action pairs. The RL agent uses this model to constrain unsafe policy updates and prioritize experiences that are unfamiliar or potentially risky. This dual mechanism aims to ensure safer, more efficient exploration and robust policy improvement, especially in challenging and dynamic environments (see Fig. 1). Our main contributions are:

- 1) A new “Feariosity” model that jointly quantifies fear and curiosity for RL-based navigation, enabling adaptive safety-exploration trade-offs.
- 2) A policy constraint method that penalizes unsafe actions while encouraging exploration of unfamiliar regions, improving both safety and learning efficiency.
- 3) An experience replay strategy that prioritizes samples involving risk and novelty, enhancing data efficiency.
- 4) Extensive validation in both simulation and real-world AGV scenarios, demonstrating consistent improvements over advanced RL and diffusion-based baselines.

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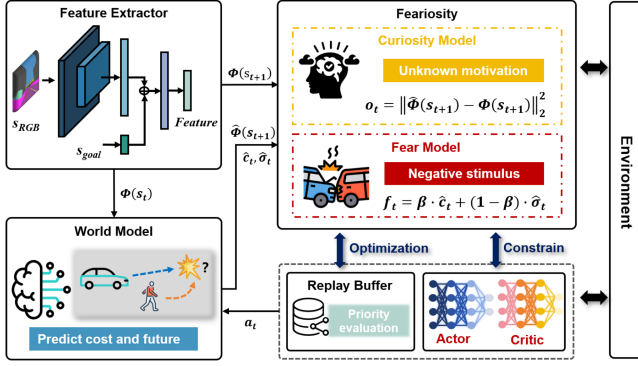


Fig. 1. The overall framework of the proposed Fog-RL: The feature extractor represents the state and extracts useful information; the world model estimates potential threats and future states; the “feariosity” model constrains and optimizes RL; and the navigation scenario’s perceptual information is semantic segmentation images.

II. RELATED WORKS

A. Safe RL for Vehicles

Safety checkers passively block risky actions, such as adding external modules to RL outputs, improving interpretability but limiting the action space and relying on expert knowledge [4], [12]. Alternatively, constrained RL algorithms enhance safety by adding constraint objectives, as seen in conservative Q-learning for car-following [13] and overtaking [14], or by incorporating risk factor into value functions [15]. Popular approaches also include Lyapunov and control barrier functions, which have been used in conflict-aware strategies [16] and stability-guaranteed methods [17]. These techniques improve Soft actor-critic (SAC)-based navigation by leveraging Lagrangian methods to maintain costs within thresholds [18] and optimize strategies using dual theory [19]. Adversarial RL objectives further enhance safety under uncertainty [20]. World models simulate trajectories or assess actions to reduce risky exploration, improve sample efficiency, and enforce cost constraints [21]. However, these methods depend on model accuracy, and excessive constraints may hinder exploration and reduce efficiency.

B. Efficient RL for Vehicles

Improving data efficiency is vital in RL applications, especially in autonomous driving. Model-based RL reduces real-world interactions and has been applied to urban traffic [22] and track scenarios [23]. Prior knowledge-based RL boost initial performance and learning speed by using pre-trained models [24] and task-specific knowledge. Additional knowledge, such as reward shaping [25], and expert experiences [26], further narrows the exploration space and enhances efficiency. Curiosity-driven RL uses intrinsic rewards to encourage exploration, improving adaptability and understanding. Techniques like inverse dynamics models [27], episodic memory [28], and random network distillation [29] design such rewards. However, this approach risks unsafe behavior in unfamiliar settings due to the absence of explicit safety constraints.

III. METHODOLOGY

A. Overview

This section introduces the proposed RL-based autonomous navigation framework based on the “Feariosity” model. As shown in Fig. 1, we first use self-supervised training to establish a feature extractor that removes irrelevant noise from the perception state. Next, a world model is developed to predict potential dangers and future states of the environment. We then construct a fear model to estimate the agent’s fear of recognizing dangers or unexpected events, and a curiosity model to stimulate the agent’s intrinsic motivation to explore the environment. Finally, we establish a **Feariosity-guided RL (Fog-RL)** algorithm, which includes Feariosity-based strategy constraints and prioritized experience replay (PER) optimization, ensuring that the agent makes safe decisions while being encouraged to explore.

B. Problem Formulation

1) *Markov Decision Process*: The AGV navigation is modeled using a constrained Markov decision process (CMDP), represented by a 6-tuple $(\mathcal{S}, \mathcal{A}, \mathcal{T}, r_t, c_t, \gamma)$ [19]. Here, t is the time step, $\mathcal{S}$ is the state space, $\mathcal{A}$ the action space, $\mathcal{T}$ is the transition model, r_t is the reward function, c_t denotes the cost function, and $\gamma \in (0, 1)$ the discount factor.

2) *State Space*: This study focuses on the motion planning performance of AGVs using a single camera input. Observations are made with three consecutive frames. RGB images are captured from the simulator, converted to single-channel grayscale images, and processed via semantic segmentation:

$$\mathcal{S} = [\Phi_{\text{seg}}([\chi_{t-2}, \chi_{t-1}, \chi_t]), d_t, \vartheta_t], \quad (1)$$

where $\chi_t \in \mathbb{R}^{H \times W}$ represents the semantic image with a height H and width W pixel matrix, and Φ_{seg} is the segmentation function. In addition, the relative distance d_t and angle ϑ_t between the ego vehicle and the destination are used to define the position state.

3) *Action Space*: The action space $\mathcal{A}$ consists of the pedal p_t and steering δ_t commands, granting the ego agent full control.

4) *Reward Function*: The objective of the AGV is to safely and efficiently reach its destination while avoiding collisions, and optimizing driving performance. Therefore, the first term of the reward function is $r_{t,\text{goal}}$, a positive constant C_1 that represents the instantaneous reward for the AGV successfully reaching its target. Conversely, if the agent collides with an obstacle, it incurs a significant penalty denoted as $r_{t,\text{collision}} = C_2$, where C_2 is a negative constant. In all other situations, both $r_{t,\text{goal}}$ and $r_{t,\text{collision}}$ are set to 0.

Then $r_{t,\text{speed}} = \omega_1 \cdot v_{t,\text{ego}}$ and $r_{t,\text{comfort}} = \omega_2 \cdot |\delta_t| \cdot (1 + v_{t,\text{ego}})$ represent that the AGV should complete the task as quickly and comfortably as possible, where $v_{t,\text{ego}}$ is speed of ego vehicle, and ω_1 and ω_2 are the weights. Thus, these sub-components form the total reward:

$$r_t = r_{t,\text{goal}} + r_{t,\text{collision}} + r_{t,\text{speed}} + r_{t,\text{comfort}}. \quad (2)$$

5) *Cost Function*: The cost function constrains the agent’s update process, focusing only on highly unreasonable situations.

Specifically, the cost c_t is defined as C_3 if a collision occurs, and 0 otherwise. This setup imposes a constant penalty, C_3 , on the AGV during a collision, discouraging unsafe behavior and emphasizing the importance of avoiding obstacles to ensure safety.

C. Fear Model

In biological organisms, fear is triggered by negative stimuli, helping address threats through learned and remembered responses like conditioned reflexes. In constrained RL, the cost function acts as a negative stimulus (e.g., a collision triggering fear). Fear also stems from cognitive uncertainty about potential threats, as unpredictability amplifies perceived risk [10]. Based on this, we designed a fear model combining the cost function and uncertainty, expressed as:

$$f(s_t, a_t) = \beta \cdot \hat{c}(s_t, a_t) + (1 - \beta) \cdot \hat{\sigma}(s_t, a_t), \quad (3)$$

where, s_t and a_t represent the current state and action. $\hat{c}(s_t, a_t)$ is the estimated cost, measuring the risk or penalty of taking action a_t in state s_t . $\hat{\sigma}(s_t, a_t)$ is the estimated uncertainty. The parameter $\beta = 0.9$ is empirically chosen to balance the effects of risk and uncertainty in the fear model. Since $\hat{c}(s_t, a_t)$ and $\hat{\sigma}(s_t, a_t)$ may have different scales, both are normalized during implementation to ensure their sum is meaningful.

To predict potential dangers in the current state, we build a world model $\mathcal{M}_\psi$ to enhance the agent's understanding and prediction of its surroundings. Unlike most conventional environment models, which predict rewards and the next state directly, our world model $\mathcal{M}_\psi$ predicts both the cost $\hat{c}_t$ and the embedding of next state $\hat{\Phi}(s_{t+1})$. It can be represented as:

$$\hat{\Phi}(s_{t+1}), \hat{c}_t \leftarrow \mathcal{M}_\psi(\Phi(s_t), a_t), \quad (4)$$

where $\Phi(\cdot)$ denotes the state embedding, which maps the high-dimensional state s_t into a lower-dimensional latent space while capturing its essential features. And we used a Gaussian distribution $\mathcal{N}(\cdot)$ as output in the cost prediction branch (c), expressed as follows:

$$\mathcal{M}_\psi^{(c)}(\Phi(s_t), a_t) = \mathcal{N}(\mu_\psi^{(c)}(\Phi(s_t), a_t), \sigma_\psi^{2(c)}(\Phi(s_t), a_t)), \quad (5)$$

where $\mu_\psi^{(c)}(\cdot)$ and $\sigma_\psi^{(c)}(\cdot)$ represent the mean and standard deviation of the Gaussian distribution. In particular, within the fear model, the estimated cost $\hat{c}(s_t, a_t)$ and uncertainty $\hat{\sigma}(s_t, a_t)$ are directly given by the mean and standard deviation of the cost predicted by the world model, i.e., $\hat{c}(s_t, a_t) = \mu_\psi^{(c)}(\Phi(s_t), a_t)$ and $\hat{\sigma}(s_t, a_t) = \sigma_\psi^{(c)}(\Phi(s_t), a_t)$.

To ensure the stability of RL learning, the training of the world model is performed at the beginning of each episode using real environment data, before the agent starts exploration. The update of this model can be achieved by minimizing the objective function based on negative log-likelihood:

$$\mathcal{L}_\mathcal{M}(\psi) = -\mathbb{E}[\log \mathcal{M}_\psi(\Phi(s_{t+1}), c_t | \Phi(s_t), a_t)]. \quad (6)$$

Therefore, in the case of a short prediction horizon k , the agent's fear can be expressed as:

$$f(s_t, a_t) = \beta \cdot \hat{c}(\hat{s}_k, a_k) + (1 - \beta) \cdot \hat{\sigma}(\hat{s}_k, a_k) \quad (7)$$

where $\hat{s}_k$ and a_k denote the predicted state and action at step k generated by k -step forward planning using the world model.

D. Curiosity Model

When facing the unknown, organisms are driven by curiosity to explore, learn, and adapt. This intrinsic motivation helps them discover resources and handle unpredictable but learnable states, addressing the issue of sparse rewards [27]. Curiosity can be modeled by using a world model to predict future states.

To filter out irrelevant features from perception states (images), we design a feature extractor that captures the truly important parts of the state. Specifically, the important parts refer to things that the agent can control and things the agent cannot control but that can affect the agent (e.g., moving obstacles). We use two submodules, as described in the following, to train the feature extractor E . The first submodule encodes the given raw state s_t into a feature vector $\Phi(s_t)$. The second submodule takes $\Phi(s_t)$ and the representation of the next state $\Phi(s_{t+1})$ as input and predicts the action $\hat{a}_t$, defined as:

$$\hat{a}_t = E(\Phi(s_t), \Phi(s_{t+1})). \quad (8)$$

It comes from supervised learning by predicting the transition process from s_t to s_{t+1} , and its loss function is $\mathcal{L}_E(\theta^E) = \|\hat{a}_t - a_t\|_2^2$. Consistent with the world model, the feature extractor is also trained at the beginning of each episode, ensuring stable feature representation throughout the episode.

The world model $\mathcal{M}_\psi$ mentioned by (5) serves as the forward dynamic model, generating $\hat{\Phi}(s_{t+1})$, while $\Phi(s_{t+1})$ is encoded by the feature extractor E . The curiosity model is defined as:

$$o(s_t, a_t, s_{t+1}) = \left\| \hat{\Phi}(s_{t+1}) - \Phi(s_{t+1}) \right\|_2^2. \quad (9)$$

The inverse model helps learn the feature space, which encodes only the information relevant to predicting the agent's actions, while the forward model makes this learned feature representation more predictable.

Based on the proposed fear ((7)) and curiosity model ((9)), the established "Feariosity" can be expressed as:

$$Fo(s_t, a_t, s_{t+1}) = f(s_t, a_t) - \Xi o(s_t, a_t, s_{t+1}), \quad (10)$$

where $\Xi = 1$ is a weight that controls the influence of curiosity. It expresses a state of mutual restraint between the agent's inner fear and curiosity when exploring the environment.

The design choice of combining fear and curiosity into a single constraint is motivated by their complementary roles in exploration. Fear, based on predicted potential risks and uncertainties, aims to constrain the agent's behavior to ensure safety. However, relying solely on fear may lead to overly conservative behaviors, limiting the agent's ability to explore and discover optimal solutions. By incorporating curiosity, which encourages exploration of unknown states, "Feariosity" counterbalances

the restrictive nature of fear, promoting a more balanced and efficient exploration process.

E. “Feariosity”-Constrained Reinforcement Learning

The “Feariosity” balances safety and exploration by combining fear-based constraints to penalize unsafe actions and curiosity to encourage exploration. Based on CMDP, Fog-RL aims to solve the following constrained optimization problems:

$$\begin{aligned} \max_{\pi} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) + \alpha H(\pi) \right], \\ \text{s.t. } \mathbb{E} [Fo(s_t, a_t, s_{t+1})] \leq Fo^* \end{aligned} \quad (11)$$

where Fo is the proposed “Feariosity”, and $Fo^* = 5$ is the specified threshold. $H(\pi)$ is the entropy of the distribution of policy’s action, denotes $\mathbb{E}[-\pi \cdot \log \pi]$. α is a regularized coefficient used to control the importance of entropy, which can be automatically adjusted by (12):

$$\mathcal{L}_{\alpha}(\alpha) = \mathbb{E} [-\alpha \log \pi(a_t | s_t) - \alpha \mathcal{H}_0], \quad (12)$$

where $\mathcal{H}_0$ is the target value of entropy. The Lagrangian of constrained optimization problems can be written as:

$$\mathcal{L}(\pi, \alpha, \lambda) = -\mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t + \alpha H(\pi) + \lambda (Fo^* - Fo) \right], \quad (13)$$

where $\lambda \geq 0$ represents the dual variable. The SAC algorithm uses two value functions, with parameters θ_i^Q and $i \in \{1, 2\}$, to prevent overestimation. The parameters of the value function can be learned by minimizing the loss function:

$$\mathcal{L}_Q(\theta_i^Q) = \mathbb{E} [\|y - Q(s_t, a_t)\|_2^2], \quad (14)$$

where $y = r + \gamma \tilde{Q}(s_{t+1}, a_{t+1})$ represents the target value, and constrained value function $\tilde{Q}$ can be designed as:

$$\tilde{Q}(s_{t+1}, a_{t+1}) = Q(s_{t+1}, a_{t+1}) - \alpha H(\pi) - \lambda Fo(s_t, a_t, s_{t+1}). \quad (15)$$

Based on [30], we use only real interaction data to train the critic network, ensuring effective training and reducing the reliance on the precision of the world model. Thus, the optimal policy of the agent can be approximated by minimizing the following objective function for the actor network:

$$\mathcal{L}_{\pi}(\theta^{\pi}) = \mathbb{E} [\alpha \log \pi + \lambda Fo(s_t, a_t, s_{t+1}) - Q^{\pi}(s_t, a_t)]. \quad (16)$$

The inclusion of $\lambda Fo(s_t, a_t, s_{t+1})$ penalizes actions that are either too risky or insufficiently exploratory.

In addition, the dual variable λ can be updated by minimizing the following objective function:

$$\mathcal{L}_d(\lambda) = \mathbb{E} [\lambda (Fo^* - Fo(s_t, a_t, s_{t+1}))]. \quad (17)$$

F. “Feariosity”-Optimized Prioritized Replay Mechanism

Traditional PER improves sample efficiency by assigning priorities to the samples in the replay buffer. For each sample, the priority is typically defined as $\rho_t = |\Delta_t^{\text{TD}}| + \epsilon$, where $\epsilon > 0$

TABLE I
PARAMETER SETTINGS FOR FOG-RL NAVIGATION

Parameter	Value
Reward weight ($C_1, C_2, C_3, \omega_1, \omega_2$)	30, -30, -10, 0.2, -0.1
Discount factor γ	0.99
Minibatch size	32
Reply buffer size	5e4
Initial entropy weight	1

TABLE II
NEURAL NETWORK SETTINGS FOR ALL COMPONENTS

Component	Parameter	Value
RL Agent	Input image shape	[64, 128, 3]
	Convolution filters	[16, 64, 256] (kernel 5×5)
	Pooling layer	Adaptive average pooling
	Goal state embedding	[32]
	Network FC layers	[256+32, 128, 32]
Feature Extractor	Convolution filters	[16, 64, 256] (kernel 5×5)
	Pooling layer	Adaptive average pooling
	Goal state embedding	[32]
	Network FC layer	[256+32, 64]
World Model	Common FC layers	[64+2, 128, 256]
	Feature branch FC layers	[256, 128, 64]
	Cost branch FC layers	[256, 64, 1]

ensures numerical stability. The temporal-difference (TD) error, Δ_t^{TD} , is calculated as:

$$\Delta_t^{\text{TD}} = r_t + \gamma \cdot Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t). \quad (18)$$

The Fog-RL proposed in this study not only introduces policy constraints but also explores how to optimize the PER mechanism. To enhance the handling of risky and unfamiliar data trajectories, the PER weights can be rewritten as follows:

$$\tilde{\rho}_t \equiv \rho_t + \exp[f(s_t, a_t) + \Xi o(s_t, a_t, s_{t+1})]. \quad (19)$$

The exponential form of the fear and curiosity functions evaluates the quality of trajectory data by prioritizing risky or novel transitions. The exponential design further amplifies differences between trajectories, assigning higher weights to the most informative ones in the PER mechanism.

The parameter settings and network structure of the proposed method are listed in Tables I and II, respectively.

IV. EXPERIMENTS

A. Simulation Configuration

To train the RL-based policy, we created a dynamic, randomized scenario in a simulated environment. The scenario, covering an enclosed area of 30×40 meters (Fig. 2), places the ego vehicle in a random initial position and orientation within a circular area of 5 m radius centered in the region. Obstacles include both static objects and dynamic pedestrians. Each episode features 20 to 30 pedestrians, 80% moving at random speeds of 0.5 to 1.5 m/s and the remaining 20% stationary. Moving pedestrians walk back and forth, reversing direction upon reaching the boundary of the area. They stop when the AGV center of mass comes within 2

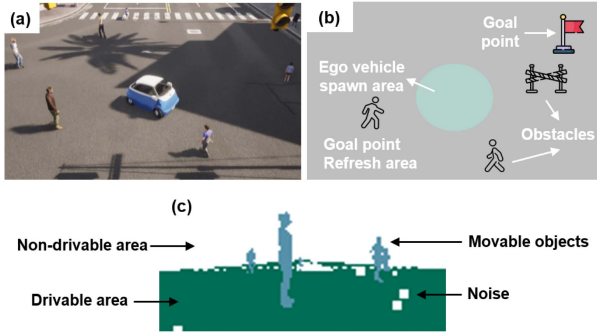


Fig. 2. Navigation task diagram in simulation. (a) Rendered scenario within the simulation environment. (b) Scenario definition, where the spawn point of the ego vehicle is randomly selected within a circular area (green), and its target position is randomly generated from the surrounding area (gray). (c) Example semantic images collected from the simulator.

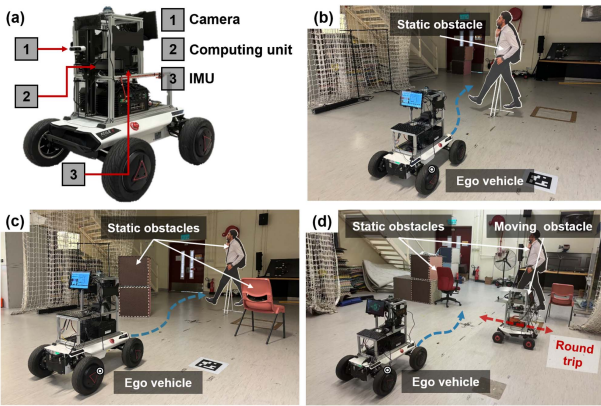


Fig. 3. Real-world testing: (a) AGV experimental platform. Scenario illustration of field tests: (b) scenario 1, (c) scenario 2 and (d) scenario 3.

meters, indicating a path overlap. The ego vehicle must reach its target without colliding with obstacles.

We built this virtual environment using the CARLA simulator. Camera sensors output segmented images with three semantic categories: drivable areas, non-drivable areas, and obstacles. The environment module of Fig. 2(c) illustrates a sample of these images. To improve policy robustness and generalization, domain randomization (DR) was applied by varying the parameters of the scenario. DR variables include randomizing the camera's installation height between 0.5 and 1.0 meters, and switching the field of view between [90, 100, 110, 120] degrees. The simulation runs at a fixed rendering rate of 10Hz, with the agent's control step at 0.1 seconds.

The training algorithm was implemented using Python scripts, with the NN model built using the PyTorch toolkit. Each policy underwent reliability training with 5 random seeds. The simulator was run on a workstation equipped with an Intel i7-13700KF CPU and an Nvidia RTX 4070 Ti GPU.

B. Real-World Configuration

Fig. 3(a) shows the AGV (HUNTER SE) used, which is equipped with a D435 camera, NVIDIA JETSON AGX Orin (64 GB) computing unit, and Inertial Measurement Unit. All

TABLE III
QUANTIFICATION OF TRAINING RESULTS

Metric	Vanilla-RL	RL-ICM	RL-Lag	Fog-RL
M. R. $\uparrow$	0.29 ± 0.03	0.32 ± 0.01	0.38 ± 0.04	0.42 ± 0.01
Succ. $\uparrow$	0.65 ± 0.03	0.72 ± 0.02	0.76 ± 0.04	0.84 ± 0.02
Coll. $\downarrow$	0.28 ± 0.07	0.22 ± 0.02	0.17 ± 0.05	0.04 ± 0.01
TNS $\uparrow$	444 ± 31	503 ± 61	511 ± 57	681 ± 29
TNC $\downarrow$	275 ± 11	237 ± 14	205 ± 42	92 ± 12

Note: M. R. is mean reward, Succ. is success rate (%), Coll. is collision rate (%). TNS and TNC are the total numbers of successes and collisions, respectively. The metric is represented as "mean $\pm$ standard deviation".

perceptions are semantically segmented using Segformer [31], with an inference frequency of 10 Hz.

We conducted real-world quantitative testing in three scenarios to evaluate the performance of different navigation strategies. Scenario 1 (Fig. 3(b)) represents a relatively simple environment where a stationary human body model is placed on the road ahead of the vehicle, and the AGV needs to take a detour. Scenario 2 (Fig. 3(c)) introduces moderate challenges with increased complexity. In addition to the human body model, other static obstacles are placed on the road, requiring the AGV to navigate carefully within a more confined driving space. Scenario 3 (Fig. 3(d)) increases the difficulty by adding a moving vehicle in front of the static obstacles. The moving obstacle maintains a constant speed for a certain period before randomly varying its speed within a small range at different intervals, requiring the AGV to adapt to dynamic changes while avoiding collisions. The target and AGV initial positions in each experiment are randomly assigned within a certain range to simulate the variability of the real world. In qualitative experiments, human participants were used to test the feasibility of the strategy.

V. RESULTS AND DISCUSSION

A. Comparison Baselines

To evaluate the proposed method, we adopt several baselines. Vanilla-RL employs the SAC algorithm [32] with identical state representations. RL-ICM [27] adds intrinsic rewards to encourage exploration and improve learning efficiency. RL-Lag [33] incorporates safety constraints into SAC, enabling safe RL.

Additionally, we test behavior cloning (BC) [34], which uses supervised learning to mimic expert actions; Diffusion-BC [35], which applies diffusion models to generate diverse actions; and Diffusion-QL [36], which integrates Q-learning with diffusion models to incorporate reward-based exploration.

B. Simulation Results

1) *Training Performance Evaluation:* Fig. 4 shows the reward, success rate, and collision rate curves during training. Table III summarizes the mean metrics from the last 100 episodes, along with the total number of successes (TNS) and collisions (TNC) throughout training. Vanilla RL, lacking effective mechanisms, shows slow improvement and the worst asymptotic performance. ICM-driven RL improves training efficiency, achieving about a 7% higher success rate than standard

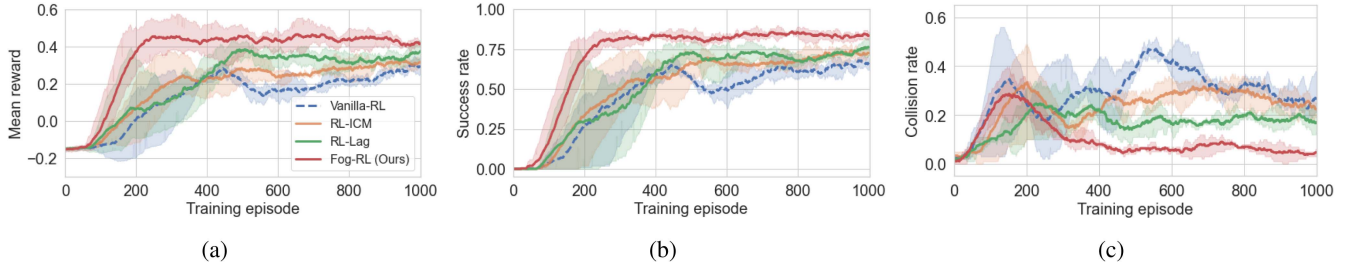


Fig. 4. Training performance of RL-based methods: (a) Mean reward; (b) Success rate; (c) Collision rate.

TABLE IV
COMPARISON OF TESTING PERFORMANCE ACROSS DIFFERENT METHODS

Method	Normal Scenario		
	Safety rate (%) $\uparrow$	Success rate (%) $\uparrow$	Time (s) $\downarrow$
Vanilla-RL	83 $\pm$ 4	79 $\pm$ 5	12.1 $\pm$ 0.9
RL-ICM	86 $\pm$ 4	84 $\pm$ 4	11.2 $\pm$ 1.1
RL-Lag	92 $\pm$ 3	85 $\pm$ 4	11.6 $\pm$ 0.4
BC	90 $\pm$ 2	82 $\pm$ 4	16.1 $\pm$ 0.6
Diffusion-BC	96 $\pm$ 1	86 $\pm$ 3	16.3 $\pm$ 1.5
Diffusion-QL	100 $\pm$ 0	94 $\pm$ 2	11.5 $\pm$ 0.9
Fog-RL	100 $\pm$ 0	97 $\pm$ 1	10.8 $\pm$ 0.3
Method	Challenging Scenario		
	Safety rate (%) $\uparrow$	Success rate (%) $\uparrow$	Time (s) $\downarrow$
Vanilla-RL	69 $\pm$ 6	65 $\pm$ 4	12.9 $\pm$ 1.1
RL-ICM	75 $\pm$ 2	74 $\pm$ 1	12.2 $\pm$ 1.2
RL-Lag	85 $\pm$ 2	66 $\pm$ 3	13.5 $\pm$ 0.2
BC	77 $\pm$ 5	70 $\pm$ 4	18.2 $\pm$ 1.3
Diffusion-BC	84 $\pm$ 4	74 $\pm$ 4	17.6 $\pm$ 1.8
Diffusion-QL	92 $\pm$ 2	85 $\pm$ 4	13.6 $\pm$ 1.6
Fog-RL	96 $\pm$ 2	87 $\pm$ 3	12.3 $\pm$ 0.8

RL due to better data utilization. Safe RL with Lagrangian constraints achieves higher asymptotic performance, reducing collision rates by about 11% in later training stages. In contrast, our proposed Fog-RL leverages a world model with a fear function for safer and more generalizable policy learning, while the curiosity model enhances exploration efficiency. This combination significantly improves learning efficiency, asymptotic performance, and stability, with TNS and TNC metrics outperforming all baselines.

2) *Testing Results Analysis*: We evaluated the performance of the navigation algorithms on three metrics: safety rate, success rate, and time to reach the destination across two scenarios with different levels of difficulty. The normal scenario (a) included 15 randomly generated pedestrians per episode, while the challenging scenario (b) included 30 pedestrians per episode, significantly increasing the challenge of navigation. Each algorithm was tested in 100 independent episodes for both the normal and challenging scenarios, with pedestrians randomly generated for each episode.

Table IV summarizes the results. Among the baseline methods, RL-ICM and RL-Lag showed improvements in safety and success rates compared to Vanilla-RL, with RL-Lag achieving the highest safety rate. However, RL-Lag's success rate dropped

TABLE V
TESTING PERFORMANCE OF ABLATION ANALYSIS IN SIMULATION SCENARIOS

Method	Normal Scenario		Challenging Scenario	
	Succ. (%) $\uparrow$	Time (s) $\downarrow$	Succ. (%) $\uparrow$	Time (s) $\downarrow$
Fog-RL	97 $\pm$ 1	10.8 $\pm$ 0.3	87 $\pm$ 3	12.3 $\pm$ 0.8
w/o Fear	85 $\pm$ 3	10.4 $\pm$ 0.8	71 $\pm$ 5	11.9 $\pm$ 1.2
w/o Curiosity	91 $\pm$ 2	12.2 $\pm$ 1.2	77 $\pm$ 4	14.1 $\pm$ 1.4
w/o Fo-cons	83 $\pm$ 4	11.0 $\pm$ 0.7	68 $\pm$ 5	11.6 $\pm$ 0.6
w/o Fo-exp	89 $\pm$ 3	12.5 $\pm$ 1.3	73 $\pm$ 4	13.1 $\pm$ 1.3
$\beta = 0.8$	95 $\pm$ 1	11.1 $\pm$ 0.5	88 $\pm$ 2	12.7 $\pm$ 0.6
$Fo^* = 1$	87 $\pm$ 3	10.6 $\pm$ 0.6	81 $\pm$ 4	11.9 $\pm$ 1.0
$Fo^* = 10$	94 $\pm$ 2	11.5 $\pm$ 0.4	85 $\pm$ 3	12.6 $\pm$ 0.6
$\Xi = 0.1$	92 $\pm$ 3	12.0 $\pm$ 0.8	82 $\pm$ 3	13.7 $\pm$ 0.8

in the challenging scenario due to its conservative behavior, causing frequent stalling and failure to reach the goal. Newly added baselines, BC, Diffusion-BC, and Diffusion-QL, demonstrated varying performance. Diffusion-BC and Diffusion-QL achieved notable improvements in success rate and safety over BC in both scenarios, with Diffusion-QL performing especially well in the challenging scenario. However, these methods incurred higher travel times compared to other algorithms. The proposed Fog-RL algorithm exhibited robust performance, balancing safety and success rate effectively. In the normal scenario, Fog-RL achieved a 100% safety rate, outperforming most baseline methods. In the challenging scenario, Fog-RL maintained a strong success rate while achieving competitive safety performance, demonstrating its ability to handle challenging navigation tasks.

3) *Ablation Analysis of Fog-RL*: We implemented some ablation schemes in the simulation environment shown in Fig. 2 by removing different components and adjusting various hyperparameters. Specifically, we removed the fear module (w/o Fear) and curiosity module (w/o Curiosity), as well as their RL improvements: policy constraints (w/o Fo-cons) and experience replay optimization (w/o Fo-exp). Policy constraints refer to the Fo model in (15) and (16), while experience replay optimization involves removing the exponential term in (19). To analyze the effect of key hyperparameters, we varied the values of " $\beta = 0.9$," " $Fo^* = 5$," and " $\Xi = 1$ " in Fog-RL.

Fig. 5 shows the ratio of the average reward of ablation candidates to Fog-RL during the same training phase, while Table V presents their performance in testing scenarios. Notably, negative values of the reward ratio occur when the ablated

TABLE VI
TESTING RESULTS IN REAL-WORLD SCENARIOS

		Vanilla-RL	RL-ICM	RL-Lag	BC	Diffusion-BC	Diffusion-QL	Fog-RL
Scenario 1	Succ. (%) $\uparrow$	15/25 (60%)	15/25 (60%)	17/25 (68%)	11/25 (44%)	18/25 (72%)	25/25 (100%)	25/25 (100%)
	Time (s) $\downarrow$	12.3 $\pm$ 0.63	11.6 $\pm$ 0.47	12.6 $\pm$ 0.58	13.4 $\pm$ 0.45	13.5 $\pm$ 0.75	13.0 $\pm$ 0.59	12.1 $\pm$ 0.41
Scenario 2	Succ. (%) $\uparrow$	5/25 (20%)	7/25 (28%)	12/25 (48%)	10/25 (40%)	15/25 (60%)	22/25 (88%)	25/25 (100%)
	Time (s) $\downarrow$	13.6 $\pm$ 0.28	13.2 $\pm$ 0.39	14.0 $\pm$ 0.50	14.5 $\pm$ 0.57	14.2 $\pm$ 0.93	13.9 $\pm$ 0.87	13.1 $\pm$ 0.56
Scenario 3	Succ. (%) $\uparrow$	6/25 (24%)	8/25 (32%)	10/25 (40%)	4/25 (16%)	8/25 (32%)	18/25 (72%)	23/25 (92%)
	Time (s) $\downarrow$	18.3 $\pm$ 0.78	18.7 $\pm$ 0.63	19.5 $\pm$ 1.03	22.9 $\pm$ 1.35	21.0 $\pm$ 1.48	21.2 $\pm$ 1.34	18.4 $\pm$ 0.69

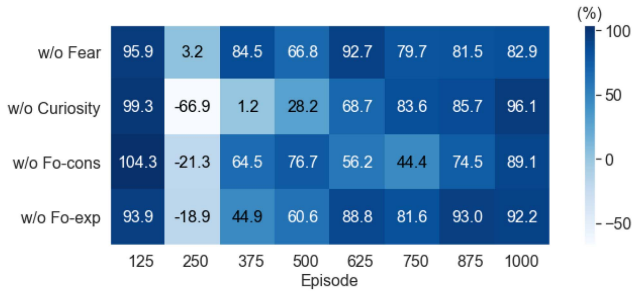


Fig. 5. Results of the ablation study, where the numbers represent the ratio of ablation candidates to the mean reward value of the proposed method during the same training phase in simulation.

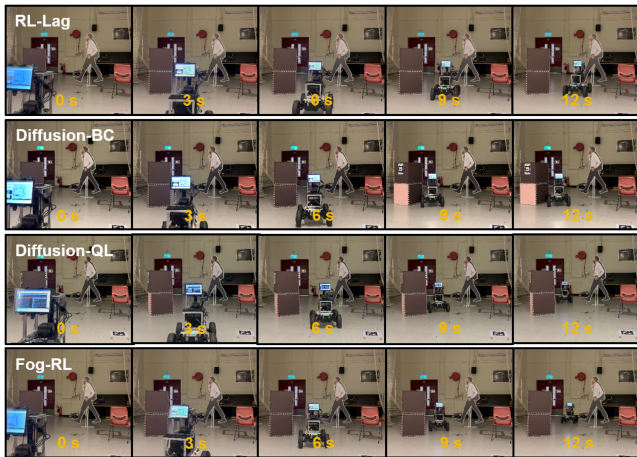


Fig. 6. Representative video frames of different agents in the experimental demons of scenario 2.

method yields negative average rewards (due to high collision rates in early training), while the full Fog-RL model achieves positive rewards in the same phase. The results show that each Fog-RL component is crucial to balance safety and exploration. Removing the fear module compromises policy safety; although initial rewards increase due to curiosity-driven exploration, the lack of constraints leads to reckless behavior, resulting in lower success rates in the later training stages and testing. Similarly, removing the curiosity module or Fo-exp severely hinders early exploration efficiency, while the absence of Fo-cons causes instability and significant training fluctuations. These factors greatly impact training performance and ultimately make it difficult for the agent to handle challenging scenario tests. For

different hyperparameter settings, the results show that while hyperparameters have some impact on Fog-RL’s performance, the method is generally robust to hyperparameter variations compared to component ablations. These results show that each component plays a key role in the learning capability of Fog-RL.

C. Real-World Experiments

We conducted real-world vehicle tests in three scenarios, evaluating each strategy with five random seed models tested five times, for a total of 25 trials. Table VI summarizes the results, including success rate and average task completion time. Vanilla RL showed low success rates across all scenarios, reflecting its limited robustness. RL-ICM, RL-Lag, and BC moderately improved success rates, though BC had the longest completion time due to reliance on static behavioral patterns. Diffusion-based methods, Diffusion-BC and Diffusion-QL, performed well in static scenarios (Scenarios 1 and 2). However, in the dynamic Scenario 3, BC and Diffusion-BC struggled, with BC achieving only a 16% success rate, the lowest among all methods, as its imitation learning mechanism lacks adaptability to dynamic obstacles, leading to collisions or getting stuck. The proposed Fog-RL method consistently performed well across all scenarios, maintaining high success rates and competitive driving efficiency. In the dynamic Scenario 3, the observed slight failure rate may be due to limitations in perception range, which occasionally leads to suboptimal reactions in rapidly changing situations. Despite this, Fog-RL demonstrated superior adaptability, ensuring efficient and reliable navigation even in the most challenging dynamic environments.

Representative experimental results are provided in part 2 of the supplementary video, and video frames from Scenario 2 are illustrated in Fig. 6. RL-Lag stalled in front of obstacles, failing to find a feasible path to avoid them, while Diffusion-BC successfully avoided pedestrians but collided with nearby objects. Both Diffusion-QL and Fog-RL successfully completed the task. Notably, the video frame at the 9 s second highlights that the proposed approach appears to plan safer and more efficient behaviors for obstacle avoidance.

Additionally, part 3 of the supplementary video provides qualitative evaluations of the feasibility of our strategy, demonstrating how the AGV effectively avoids moving pedestrians in dynamic environments. This showcases the robustness of the proposed method in handling real-time interactions with unpredictable obstacles.

VI. CONCLUSION

We propose Fog-RL, a RL algorithm that integrates fear and curiosity models, designed to optimize end-to-end navigation strategies for AGVs. The “Feariosity” model introduces safety constraints and optimizes replay priority. In dense crowd navigation tasks, results demonstrate that Fog-RL outperforms other baseline methods by significantly improving safety and accelerating exploration efficiency. Additionally, real-world experiments further validate the advantages and deployment potential of this method.

However, the integration of fear and curiosity remains simplistic, with fixed weighting coefficients that may limit adaptability in dynamically changing environments. Furthermore, while this method is specifically designed for AGV navigation, its application to other RL scenarios, such as quadruped locomotion or drone trajectory planning, remains an open question. Future work will explore these extensions, as well as more adaptive mechanisms for balancing fear and curiosity to improve scalability and efficiency.

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